Total volume = 0.250L

$$\underset{0.1\ M}{Pb(NO_3)_2} \rightleftharpoons \underset{0.1\ M}{Pb^{2+}} + \underset{0.2\ M}{2NO_3^-}$$

Number of moles

Pb^{2+} = molarity × Volume of the solution in lit

$= 0.1 \times 0.15$

$$[Pb^{2+}]_{mix} = \frac{0.1 \times 0.15}{0.25} = 0.06\ M$$

$$\underset{0.2\ M}{NaCl} \rightleftharpoons \underset{0.2\ M}{Na^+} + \underset{0.2\ M}{Cl^-}$$

No.of moles $Cl^- = 0.2 \times 0.1$

$$[Cl^-]_{mix} = \frac{0.2 \times 0.1}{0.25} = 0.08\ M$$

Precipitation of $PbCl_2$ (s) occurs if

$[Pb^{2+}][Cl^-]^2 > K_{sp}$

$[Pb^{2+}][Cl^-]^2 = (0.06)(0.08)^2$

$= 3.84 \times 10^{-4}$

Since ionic product $[Pb^{2+}][Cl^-]^2 > K_{sp}$, $PbCl_2$ is precipitated

27. $Al(OH)_3 \rightleftharpoons Al^{3+}(aq) + 3OH^-(aq)$

$K_{sp} = [Al^{3+}][OH^-]^3$

$Al(OH)_3$ precipitates when

$[Al^{3+}][OH^-]^3 > K_{sp}$

$(1 \times 10^{-3})[OH^-]^3 > 1 \times 10^{-15}$

$[OH]^3 > 1 \times 10^{-12}$

$[OH^-] > 1 \times 10^{-4} M$

$[OH^-] = 1 \times 10^{-4} M$

$POH = -\log_{10}[OH^-] = -\log(1 \times 10^{-4}) = 4$

pH = 14-4= 10

Thus, $Al(OH)_3$ precipitates at a pH of 10

Evaluate yourself

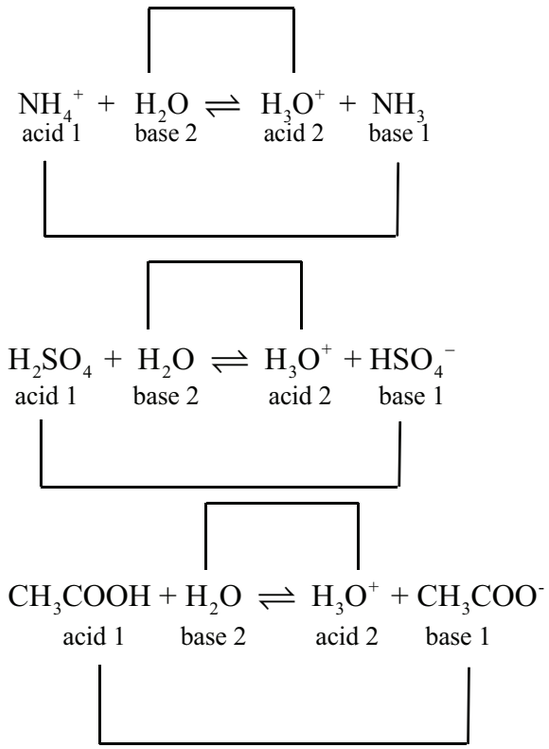
Key

Evaluate yourself – 1

acid : (i) HNO_3 iii) H_3PO_3 iv) CH_3COOH

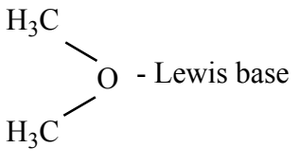
base : ii) $Ba(OH)_2$

Evaluate yourself – 2



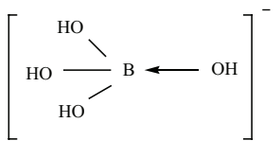
Evaluate yourself – 3

i) CaO - Lewis base ; CO_2 -Lewisacid

ii)  - Lewis base

$AlCl_3$ - Lewis acid

Evaluate yourself – 4

 ; electron pair acceptor - Lewis acid

Evaluate yourself – 5

Given solution is neutral

$\therefore [H_3O^+] = [OH^-]$ Let $[H_3O^+] = x$; then $[OH^-] = x$

$K_w = [H_3O^+][OH^-]$

$4 \times 10^{-14} = x \cdot x$

$x^2 = 4 \times 10^{-14}$

$x = \sqrt{4 \times 10^{-14}} = 2 \times 10^{-7}$

Answers.indd 303 16-12-2022 19:28:58